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POWER BI NOTES FOR BEGINNERS

Comprehensive Reference Guide: Architecture, Power Query, Data Modeling, DAX, Visualizations, Dashboards, and Power BI Service (Topics 1 to 86)

1. What is Power BI?

Power BI is a business intelligence and data visualization tool developed by Microsoft. It helps users connect to data, clean and transform it, create interactive reports, and share insights.

Key points:

- Connects data from many sources
- Transforms and organizes data
- Creates charts, tables, and dashboards
- Helps make data-driven decisions

2. Why do we use Power BI?

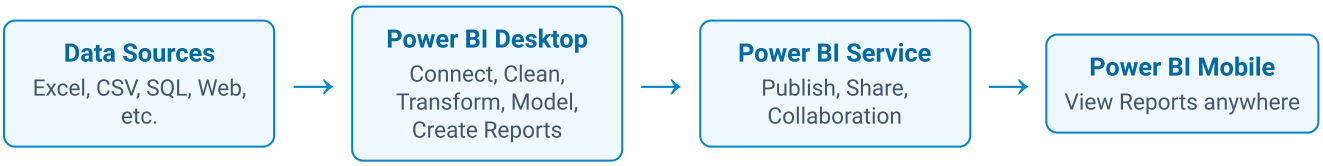
Power BI converts raw data into meaningful information. It makes it easier to understand trends, compare results, track performance, and present findings visually.

3. Main Components of Power BI

- **Power BI Desktop:** builds data models and reports
- **Power BI Service:** publishes and shares reports online
- **Power BI Mobile:** views reports on phones and tablets

4. How does Power BI Work? (Architecture)

Power BI works in a simple sequential flow:



Explanation:

- **Data Sources:** Power BI can connect to many data sources like Excel, CSV, SQL Server, Web, etc.
- **Power BI Desktop:** Used to clean, transform, model data and create reports.
- **Power BI Service:** Cloud platform to publish and share reports.
- **Power BI Mobile:** Mobile app to view reports on phones and tablets.

5. Versions of Power BI

Power BI is available in different versions:

- **Power BI Desktop:** Free, used for report creation (Windows).
- **Power BI Pro:** Paid, per user license for sharing and collaboration.
- **Power BI Mobile:** Mobile app for viewing reports.
- **Power BI Report Server:** For on-premises reporting (organization use).
- **Power BI Cloud:** Cloud platform for sharing and collaboration.
- **Power BI Premium:** Advanced features for large organizations.

6. Features of Power BI

- Easy to connect with multiple data sources
- Powerful data transformation with Power Query Editor
- Interactive and visually appealing reports and dashboards
- Drag and drop interface (no coding required)
- Supports real-time data
- Custom visualizations with a wide range of chart options
- Data modeling and DAX for advanced calculations

7. How to Install Power BI Desktop?

- Go to official website: `https://powerbi.microsoft.com`
- Click on "Download Power BI Desktop"
- Select your operating system (Windows)
- Download and install the setup file
- After installation, open Power BI Desktop and start using it
- It is completely free for personal use

8. Power BI Desktop Interface

When we open Power BI Desktop, we get a simple and user-friendly interface with four key structural zones:

- **Menu Bar & Ribbon (Top):** Home, Insert, Modeling, View, Optimize, and Help tabs containing all action tools.
- **Report View / Canvas (Center):** The main design canvas where charts, cards, and visuals are placed.
- **Left Views Bar:** Switch between Report View, Table Data View, and Model View.
- **Panels (Right):** Filters pane, Visualizations pane (to select and format charts), and Data/Fields pane (tables and columns).

9. Different Views in Power BI Desktop

- **Report View:** Used to create charts, graphs, and dashboards.
- **Data View:** Used to view and modify raw data in table format.
- **Model View:** Used to create relationships between different tables.

10. Important Areas in Interface

- **Menu Bar / Ribbon:** Contains File, Home, Insert, Modeling, View, Optimize, Help. Provides all main tools and options.
- **Visualizations Pane:** Used to add charts and customize visuals.
- **Fields / Data Pane:** Shows all tables and columns from the data.
- **Report Canvas:** Main space to design and view reports and dashboards.

11. Connecting Data Sources in Power BI

Power BI can connect to many types of data sources like Excel, CSV, SQL Server, Web, and more. We can import data or use direct queries depending on our requirement.

Steps to Connect a Data Source:

- Open Power BI Desktop.
- Click on Home tab → Get Data.
- Choose the data source (Excel, CSV, SQL Server, Web etc.).
- Provide the file path or server details.
- Select the required tables or sheets.
- Click on **Load** (to import data) or **Transform Data** (to clean data).

12. Connecting Excel

- Click on **Get Data** → **Excel**.
- Browse and select the Excel file.
- Select the sheet or table.
- Click on **Load** to import the data.

> **Note:** Power BI works best when the Excel data is in table format.

13. Connecting CSV / Text

- Click on **Get Data** → **Text/CSV**.
- Select the CSV file from your computer.
- Check delimiter and review the columns.
- Click on **Load** to import the data.

> **Note:** CSV files are useful for simple data and are widely used for data analysis.

14. Connecting SQL Server

- Click on **Get Data** → **SQL Server**.
- Enter Server name and Database name.
- Choose the required tables.
- Click on **Load** to import the data.

> **Note:** SQL Server is commonly used in companies for large data and real-time reporting.

15. Other Data Sources in Power BI

- **Web:** Connect to data from a web page or API.
- **Text File:** Read data from plain text files.
- **Folder:** Import and combine multiple files from a single folder.

16. Power Query Editor in Power BI

Power Query Editor is a powerful tool in Power BI used for data cleaning, transformation, and shaping. It allows us to modify data before loading it into the data model.

Key features:

- Clean and transform data without writing code.
- Remove unnecessary rows and columns.
- Handle missing values and duplicates.
- Change data types.
- Merge and append queries.
- Creates custom columns.
- Use M language for advanced transformations.

17. How to Open Power Query Editor?

- Open Power BI Desktop.
- Click on **Home tab** → **Transform Data** (or Edit Queries).
- Power Query Editor window will open.
- Now you can perform all data cleaning and transformation tasks.

18. Common Data Cleaning Operations

Operation	Description
Remove Duplicates	Removes duplicate rows from the table.
Remove Blank Rows	Removes rows with null or blank values.
Keep / Remove Columns	Selects required columns or removes unwanted columns.
Rename Columns	Changes column names for better understanding.
Change Data Type	Sets correct data type (Text, Number, Date, etc.).
Replace Values	Replaces specific values with new values.
Fill Missing Values	Fills null values using default or custom value.
Split Column	Splits a column into multiple columns.
Merge Queries	Combines two tables using a common column.
Append Queries	Adds rows from one table to another table.
Create Custom Column	Creates a new column using a formula.
Use M Language	Advanced transformations using M code.

19. Example: Cleaning Sales Data

- Import Sales Data with empty and duplicate rows.
- **Remove duplicate rows.**
- **Change the data type** of Date column to Date.
- **Replace blank values** in Sales column with 0.
- Click on **Close & Apply** to load clean data.

20. Data Modeling in Power BI

Data modeling is the process of organizing data tables and creating relationships between them. A good data model helps in better analysis, faster performance, and accurate results in Power BI.

21. Important Terms in Data Modeling

- **Table:** Collection of related data in rows and columns.
- **Primary Key:** Unique value that identifies each record in a table.
- **Foreign Key:** Links two tables together.
- **Relationship:** Connection between two tables based on common columns.
- **Cardinality:** Defines how many records in one table are related to records in another table (One-to-One, One-to-Many, Many-to-Many).
- **Data Model:** Combination of tables, relationships, and calculations used for analysis.

22. Types of Relationships

- **One-to-One (1:1):** Each record in one table is related to one record in another table.
- **One-to-Many (1:*):** One record in a table can be related to multiple records in another table.
- **Many-to-Many (*:*):** Multiple records in one table can be related to multiple records in another table.

23. Example: Sales Data Model

Here we have three tables: Customers, Products, and Sales.

Customers Table

CustomerID (PK)	Name	City
1	Rahul	Delhi
2	Priya	Mumbai
3	Aman	Jaipur
4	Neha	Kolkata

Sales Table

SalesID	CustomerID (FK)	ProductID (FK)	Quantity	Amount
101	1	P1	2	500
102	2	P2	1	300
103	1	P3	5	1250
104	3	P1	1	250

Products Table

ProductID (PK)	Product Name	Price
P1	Laptop	25000
P2	Mouse	300
P3	Keyboard	500
P4	Monitor	7000

24. Relationship Diagram



25. How to Create Relationships in Power BI

- Open Power BI Desktop → Click on **Model View**.
- Drag one table column and drop it on the related table column.
- Power BI will automatically detect and create the relationship.

26. Data Visualization in Power BI

Data visualization in Power BI helps us to represent data in graphical form. It makes data easy to understand and helps in finding insights quickly. Power BI provides a wide range of types of visualizations like bar chart, line chart, pie chart, table, card, slicer, etc.

27. Types of Visualizations

Visualization	Description	Best Used For
1. Bar Chart	Used to compare values between categories. Can be vertical or horizontal.	Comparing product sales, regional metrics.
2. Column Chart	Shows data in vertical columns.	Useful for comparing different categories clearly.
3. Line Chart	Used to show trend over time. Commonly used for sales analysis.	Time-series, quarterly growth, monthly patterns.
4. Pie Chart	Shows part-to-whole relationship.	Useful when we have few categories (≤ 5).
5. Table	Displays data in rows and columns. Useful for detailed data viewing.	Audit data, itemized transaction logs.
6. Card	Shows a single numeric value in a card format (e.g. Total Sales: 12,450).	KPIs, total revenue, customer count.
7. Slicer	Used to filter data interactively on the report canvas.	Enabling dynamic filtering by Category, Year, Region.
8. Map Chart	Used to show data on a geographical map.	Location-based analysis, state/city metrics.

28. How to Add a Visualization?

- Open Power BI Desktop and go to **Report View**.
- Select the required visualization from the **Visualizations pane**.
- Drag and drop the required fields/columns into the visualization fields.
- Power BI will automatically create the chart.
- You can customize the chart using the **Format pane**.

29. Customizing Visualizations

- Change the title, color, size, and labels.
- Add data labels to show values directly on bars or points.
- Use filters and slicers for better interactivity.

30. Filters in Power BI

Filters help us to display specific data in our reports. Power BI has three main types of filters used in the Filters pane:

Filter Level	Scope	Usage
1. Visual Level Filter	Applies only to a single visual on the canvas.	Shows only the data for that specific visual.
2. Page Level Filter	Applies to all visuals on the current page.	Useful when we want to filter the entire page (e.g., Year = 2024).
3. Report Level Filter	Applies to all pages in the entire report.	Useful for global filters like Region, Year, etc.

31. Slicers in Power BI

Slicers provide an interactive way to filter data. Users can easily select one or multiple values to filter the visuals on the report page.

Benefits of Slicers:

- Easy and user-friendly visual filtering.
- Makes reports interactive for business users.
- Can be used for single or multiple selection.
- Works with all related visuals on the page.

32. Types of Slicers

- **List Slicer:** Shows values in a list with checkboxes.
- **Dropdown Slicer:** Shows values in a compact dropdown menu.
- **Date Slicer:** Used for date or time fields (e.g., Year, Quarter, Month, slider).
- **Numeric Range Slicer:** Used for number ranges (e.g., Sales amount slider).

33. How to Add a Slicer?

- Go to Report View in Power BI.
- Click on the **Slicer visual** from the Visualizations pane.
- Drag and drop the required field into the Slicer.
- Choose the slicer type from the format settings.
- Adjust the size, style, and settings as per your report design.

34. Slicer Settings and Options

- **Single Select or Multi-Select:** Allows one or multiple values to be chosen.
- **"Select All" Option:** Quickly select all values with one click.

35. Introduction to DAX in Power BI

DAX (Data Analysis Expressions) is a formula language used in Power BI for advanced calculations. It allows users to create calculated columns, measures, and custom calculations for better insights.

Key points:

- DAX is similar to Excel formulas but more powerful.
- Used to create custom measures and calculated columns.
- Helps in data aggregation, filtering, and time intelligence.
- DAX formulas are case-insensitive.
- DAX functions are used with tables and columns.

36. Types of DAX Calculations

Calculated Column	Measure	Calculated Table
• Created in Data view. • Calculated row by row. • Consumes space in the table. • Used for static calculations. **Example:** `TotalPrice = Products[Price] * Products[Quantity]`	• Created in Report view. • Calculated on the fly. • Does not store data (low memory). • Used for dynamic calculations in visuals. **Example:** `Total Sales = SUM(Sales[Amount])`	• Created using DAX formula. • Stores a new table. • Useful for creating custom tables. • Used for advanced analysis. **Example:** `TopCustomers = TOPN(10, Customers, [Total Sales])`

37. Common DAX Functions

Function	Description	Example	Result
SUM()	Adds all numeric values	`SUM(Sales[Amount])`	Total sum
COUNT()	Counts number of rows	`COUNT(Sales[SalesID])`	Total count
AVERAGE()	Returns average value	`AVERAGE(Sales[Amount])`	Average amount
MIN()	Returns minimum value	`MIN(Sales[Amount])`	Minimum amount
MAX()	Returns maximum value	`MAX(Sales[Amount])`	Maximum amount
DISTINCTCOUNT()	Counts unique values	`DISTINCTCOUNT(Customers[CustomerID])`	Unique customer count
CALCULATE()	Evaluates expression in modified context	`CALCULATE(SUM(Sales[Amount]), Sales[Category] = "Laptop")`	Sum for Laptop category

38. Examples of DAX Measures

KPI	DAX Formula	Explanation
Total Sales	`Total Sales = SUM(Sales[Amount])`	Adds all sales amount.
Total Quantity	`Total Quantity = SUM(Sales[Quantity])`	Adds all quantities.
Total Orders	`Total Orders = COUNT(Sales[SalesID])`	Counts total number of orders.
Total Customers	`Total Customers = DISTINCTCOUNT(Customers[CustomerID])`	Counts unique customers.
Max Sales	`Max Sales = MAX(Sales[Amount])`	Returns highest sales amount.
Min Sales	`Min Sales = MIN(Sales[Amount])`	Returns lowest sales amount.

40. DAX Operators in Power BI

DAX supports different types of operators which are used to perform calculations, comparisons and logical operations.

1. Arithmetic Operators

Operator	Name	Example	Result (if A = 10, B = 5)
+	Addition	`A + B`	15
-	Subtraction	`A - B`	5
*	Multiplication	`A * B`	50
/	Division	`A / B`	2
^	Exponentiation	`A ^ B`	100000
MOD	Modulo (Remainder)	`MOD(A, B)`	0

2. Comparison Operators

Operator	Meaning	Example	Result (if A = 10, B = 5)
=	Equal to	`A = B`	FALSE
<>	Not equal to	`A <> B`	TRUE
>	Greater than	`A > B`	TRUE
<	Less than	`A < B`	FALSE
>=	Greater than or equal to	`A >= B`	TRUE
<=	Less than or equal to	`A <= B`	FALSE

3. Logical Operators

Operator	Name	Example	Description
&&	Logical AND	`AND(A > 5, B < 10)`	Returns TRUE if both conditions are true.
||	Logical OR	`A > 5 || B > 10`	Returns TRUE if at least one condition is true.
NOT	Logical NOT	`NOT(A > 5)`	Returns the opposite result.

41. DAX with Conditions

DAX offers powerful conditional logic functions:

- **IF():** Used to return different values based on a condition.
- **SWITCH():** Used when we have multiple conditions.

IF Function Example

```
Status = IF(Sales[Amount] > 1000,
"High", "Low" ) Bonus =
IF(Sales[Amount] > 5000,
Sales[Amount] * 0.05, 0 )
```

Explanation: If Sales amount is greater than 1000 then show "High" else "Low". Gives 5% bonus if sales is greater than 5000, otherwise 0.

SWITCH Function Example

```
Category Type =
SWITCH(Products[Category],
"Electronics", 1, "Clothing", 2, 3 )
```

Explanation: Returns different numeric category type based on product name.

42. Practical Examples using Operators

Calculation	DAX Formula	Description
Total Cost	`SUM(Sales[Quantity] * Sales[Price])`	Multiplication using * operator.
Profit	`SUM(Sales[Amount]) - SUM(Sales[Cost])`	Subtraction using - operator.
Profit %	`DIVIDE([Profit], [Total Sales]) * 100`	Division and multiplication.
Discount	`Sales[Price] - (Sales[Price] * 0.10)`	Percentage subtraction.

43. Understanding Filter Context in DAX

Filter context means the set of filters applied to the data while calculating a formula. Filter context can come from slicers, page filters, visual filters or from DAX formulas. The result of a DAX measure changes based on the applied filter context.

44. Row Context vs Filter Context

Row Context	Filter Context
• Works row by row. • Used in calculated columns. • Refers to current row of the table. • Does not automatically filter other tables. • **Example:** `Sales[Quantity] * Sales[Price]`	• Works on a set of rows. • Used in measures. • Comes from slicers, filters, or CALCULATE(). • Filters the data before calculation. • **Example:** `SUM(Sales[Amount])` with a Category filter.

45. CALCULATE() Function

`CALCULATE()` is the most powerful DAX function. It allows us to modify the filter context and calculate a result based on new filters.

```
Syntax: CALCULATE(Expression, Filter1, Filter2, ...)
```

Key Points:

- Changes the filter context.
- Can add, remove or modify filters.
- Used mostly in measures.
- Supports logical conditions and functions.

46. Examples of CALCULATE()

DAX Formula	Description	Result / Use
`Total Sales = CALCULATE(SUM(Sales[Amount]))`	Calculates total sales amount.	Basic calculation.
`Laptop Sales = CALCULATE(SUM(Sales[Amount]), Products[Category] = "Laptop")`	Calculates sales for only Laptop category.	Applies a filter on Category.
`Delhi Sales = CALCULATE(SUM(Sales[Amount]), Customers[City] = "Delhi")`	Calculates sales for Delhi city.	Applies a filter on City.
`High Value Sales = CALCULATE(SUM(Sales[Amount]), Sales[Amount] > 1000)`	Calculates sales where amount is greater than 1000.	Uses a condition in filter.
`All Customers Sales = CALCULATE(SUM(Sales[Amount]), ALL(Customers))`	Ignores the filter on Customers table and calculates total sales.	Removes filters using ALL().

47. Other Useful Filter Functions with CALCULATE()

Function	Description	Example
ALL()	Removes all filters from a table or column.	`CALCULATE(SUM(Sales[Amount]), ALL(Products))`
FILTER()	Filters a table based on a condition.	`CALCULATE(SUM(Sales[Amount]), FILTER(Sales, Sales[Amount] > 5000))`

48. Time Intelligence in Power BI (DAX)

Time Intelligence functions in DAX are used to perform calculations on time-related data. They are used to calculate YTD (Year-to-Date), MTD (Month-to-Date), QTD (Quarter-to-Date), previous year, and period comparisons. These functions work with a proper Date table.

49. Important Points about Time Intelligence

- Time Intelligence functions work only with a Date / Calendar table.
- The Date table must have a continuous range of dates.
- Mark the date table as a "Date table" in Power BI.
- These functions automatically handle the filter context for time-based calculations.
- Common functions include: `TOTALYTD`, `SAMEPERIODLASTYEAR`, `DATESINPERIOD`, etc.

50. Creating a Date Table

Using DAX to Create Date Table

```
Date = CALENDAR( DATE(2022,1,1), DATE(2026,12,31) ) Date = CALENDARAUTO()
```

Steps to Use Date Table

- Create a new table using DAX.
- Ensure the table has a continuous date range.
- Go to Table tools → Mark as date table.
- Choose the Date column.
- Use this table in relationship with your data table.

51. Common Time Intelligence Functions

Function	Description	Syntax	Use Case
TOTALYTD()	Calculates Year-to-date value.	`TOTALYTD(expression, Date[Date])`	Total sales till current date in the year.
TOTALMTD()	Calculates Month-to-date value.	`TOTALMTD(expression, Date[Date])`	Total sales till current date in the month.

Function	Description	Syntax	Use Case
TOTALQTD()	Calculates Quarter-to-date value.	`TOTALQTD(expression, Date[Date])`	Total sales till current date in the quarter.
SAMEPERIODLASTYEAR()	Returns the same period from previous year.	`SAMEPERIODLASTYEAR(Date[Date])`	Compare with last year same period.
DATEADD()	Shifts dates by a specified time period.	`DATEADD(Date[Date], -1, YEAR)`	Can shift -1 or +1 year / month / quarter.
DATESINPERIOD()	Returns a range of dates for a specific period.	`DATESINPERIOD(Date[Date], MAX(Date[Date]), -3, MONTH)`	Get last 3 months data.
DATESBETWEEN()	Returns dates between two dates.	`DATESBETWEEN(Date[Date], StartDate, EndDate)`	Get data between two dates.

52. Practical Examples

Calculation	DAX Formula	Explanation
Total Sales (YTD)	`Total Sales YTD = TOTALYTD(SUM(Sales[Amount]), Date[Date])`	Calculates total sales from start of the year till current date.
Last Year Sales	`Last Year Sales = CALCULATE(SUM(Sales[Amount]), SAMEPERIODLASTYEAR(Date[Date]))`	Calculates sales amount for the same period in previous year.
Total Sales (QTD)	`Total Sales QTD = TOTALQTD(SUM(Sales[Amount]), Date[Date])`	Calculates total sales for current quarter.
Total Sales (MTD)	`Total Sales MTD = TOTALMTD(SUM(Sales[Amount]), Date[Date])`	Calculates total sales for current month.
Last Month Sales	`Last Month Sales = CALCULATE(SUM(Sales[Amount]), DATEADD(Date[Date], -1, MONTH))`	Gets sales for previous month.

53. Dashboard Design in Power BI

A dashboard is a single-page report that shows the most important information in a visual and easy-to-understand format. A good dashboard helps users to quickly analyze data and make better decisions.

Key Points:

- A dashboard should be simple, clear and interactive.
- Use relevant charts, KPIs and slicers.
- Focus on important metrics.
- Use consistent colors and proper layout.
- Make the dashboard easy to read and understand.

Elements of a Dashboard:

- KPI Cards (Key metrics)
- Charts and Visualizations
- Slicers and Filters
- Titles and Labels
- Theme and Colors
- Proper Layout and Alignment

54. Steps to Create a Dashboard

1. Load and prepare the data.
2. Create required measures using DAX.
3. Add KPI cards for key metrics (e.g. Total Sales, Total Profit, Total Customers).
4. Add appropriate charts (bar, line, pie, etc.).
5. Insert slicers for interactive filtering.
6. Arrange visuals in a clean and logical layout.
7. Add clear titles and format the visuals.
8. Apply a theme for a professional look.

Best Practices:

- Keep the design simple and clean.
- Use relevant and meaningful visuals.
- Use consistent color scheme.
- Avoid too much text.
- Use slicers for better interactivity.
- Highlight key metrics using KPI cards.
- Maintain proper spacing and alignment.
- Use icons and modern themes.
- Make the dashboard mobile-friendly (if needed).

55. Example of a Power BI Dashboard

Sales Analysis Dashboard (Year: 2026)

Total Sales	Total Profit	Total Orders	Total Customers
₹ 12,45,000	₹ 2,87,000	1,245	892
↑ 12%	↑ 8%	↑ 15%	↑ 5%

Main Visual Components:

- **Sales by Month:** Column/Line chart tracking Jan to Dec monthly trajectory.
- **Sales by Category:** Donut chart (Electronics 35%, Clothing 25%, Furniture 20%, Grocery 15%, Others 5%).
- **Sales by Region:** Map or horizontal bar chart (North, South, East, West).
- **Top 5 Products:** Bar chart showing best-selling revenue generators (Laptop, Mobile, TV, Fridge, Headphone).

56. Useful Tips for Professional Dashboards

Do's:

- Use consistent fonts and colors.
- Choose the right chart for the right data.
- Keep margins and spacing balanced.

Don'ts:

- Avoid clutter and unnecessary visuals.
- Avoid too many bright colors.
- Don't place too many cards or slicers.

57. Reports vs Dashboards in Power BI

Both reports and dashboards are used to visualize data in Power BI, but they serve different purposes. Reports provide detailed information with multiple pages and visual interactions, while dashboards provide a high-level summary on a single screen.

58. Key Differences: Reports vs Dashboards

Feature	Power BI Report	Power BI Dashboard
Purpose	Detailed analysis of data	High-level summary of key data
Data Source	Single dataset or data model	Can use multiple reports (published)
Pages	Can have multiple pages	Single page view
Interactivity	More interactive with filters and visuals	Limited interactivity (mostly slicers)
Customization	Highly customizable	Simple and clean layout
Visuals	Wide range of charts and visuals	Contains key visuals and tiles
Use Case	Detailed analysis, exploration	Quick insights, monitoring KPIs
Audience	Analysts, managers	Executives, top management
Created In	Power BI Desktop	Power BI Service (Online)

59. Example: Power BI Report (Detailed Analysis)

Report Structure: Multiple analytical pages (Product Analysis, Sales by Region, Monthly Trends).

Contains: Detailed breakdown tables, extensive slicers by date/category, Drill-through capabilities, deep tabular sales figures.

60. Example: Power BI Dashboard (High-Level Summary)

Dashboard Structure: Single-screen executive layout with tiles pinned across different reports.

Contains: High-level Total Sales, Profit Trend line, Regional distribution bar, Top 5 Products snapshot.

61. When to Use Reports and Dashboards?

- **Use Reports:** When you need detailed and page-wise analysis, drill-through, and data exploration.
- **Use Dashboards:** When you need a quick summary of key metrics on a single screen for executive review.

62. What is Power BI Service?

Power BI Service is a cloud-based platform provided by Microsoft. It allows us to publish reports from Power BI Desktop, share them with other users, create dashboards, manage data refreshes, and collaborate with team members. We can access Power BI Service from any web browser at `https://app.powerbi.com`.

63. Key Features of Power BI Service

- Publish and share reports online.
- Create and share dashboards.
- Schedule data refresh automatically.
- Collaborate with team members.
- Set permissions and access control.
- Access reports from anywhere using a web browser.
- Connect data sources.
- Create apps to share multiple reports and dashboards.

Feature	Power BI Desktop	Power BI Service
Platform	Desktop Application	Cloud (Web)
Purpose	Create & edit reports	Share, collaborate, manage
Data Storage	Local (.pbix)	Cloud storage
Sharing	Limited	Easy sharing with users
Data Refresh	Manual (local)	Scheduled refresh
Access	Installed on PC	Any browser (online)

64. What is a Workspace?

A workspace is a container in Power BI Service where we can store, manage and share reports, dashboards, datasets and dataflows. It helps in collaboration by allowing multiple users to work on the same project.

Types of Workspaces

- **My Workspace:** Default workspace for individual users.
- **Shared Workspace:** Used to collaborate with specific users.
- **Workspace (Collection):** Used for departments or teams.

Steps to Create a Workspace

1. Go to Power BI Service.
2. Sign in with your Microsoft account.
3. Click on "Workspaces" in the left menu.
4. Click on "Create a workspace".
5. Enter a name and description.
6. Choose workspace type (Basic, Pro, Premium).
7. Click on "Save".

65. Publishing a Report to Power BI Service

Steps to Publish a Report

1. Create your report in Power BI Desktop.
2. Click on "Publish" button in the ribbon.
3. Sign in with your Microsoft account.
4. Select the workspace.
5. Click on "Select" and wait for the upload to complete.
6. Open Power BI Service and check the published report.

After Publishing

- The report will be available in the selected workspace.
- We can create a dashboard from the report.
- We can share it with other users.
- We can set up scheduled data refresh.
- We can manage permissions.

66. Sharing Reports and Dashboards

Steps to Share a Report: Go to the report in Power BI Service → Click on "Share" button → Enter the email addresses of the users → Select permission level → Click on "Grant Access".

Permission	Description
Read	Users can only view the report and dashboard.
Viewer	Can view report and dashboard.
Contributor / Member	Can view and edit reports and data.

67. What is Power BI Gateway?

Power BI Gateway is a software bridge between on-premises data sources and Power BI Service. It allows Power BI Service to securely access and refresh data in your local network, such as SQL Server, Excel files, or other databases.

Key Points:

- Acts as a bridge between local data and cloud.
- Used for scheduled data refresh.
- Keeps data connection secure.
- Runs in the background.
- Required for on-premises data sources.
- Supports multiple data sources.

When do we need a Gateway?

- When data source is in local network (e.g. Local SQL Server, Excel, Oracle).
- For scheduled refresh of on-premises data.
- When using DirectQuery to local databases.
- To keep data secure without exposing it directly to the internet.

68. Types of Power BI Gateway

Type	Description	Use Case
Personal Mode	Used by a single user. Easy to install and can only be used by one user at a time.	For personal use or small projects.
Standard Mode	Used by multiple users in an organization. Installed on a server and managed centrally.	For organization and team use. Supports high availability and better control.

69. Steps to Install and Configure Gateway (Standard Mode)

1. Download Gateway from Microsoft website.
2. Install it on a Windows server (always on).
3. Sign in with your Microsoft account.
4. Register the gateway with your organization.
5. Create a new data source (e.g. SQL Server).
6. Enter connection details and test the connection.
7. Save and close.

Configuration Points:

- Server must have a stable internet connection.
- The server should always be running.
- Provide proper credentials to access data source.
- You can add multiple data sources.
- Manage and monitor the gateway from Power BI Service.

70. Scheduled Data Refresh in Power BI

Scheduled refresh allows Power BI Service to automatically update the data in reports and dashboards at regular intervals.

Steps to Schedule Refresh:

1. Publish the report to Power BI Service.
2. Go to the dataset settings.
3. Click on "Scheduled refresh".
4. Turn on scheduled refresh.
5. Set the refresh frequency (e.g. daily, hourly).
6. Choose the time and time zone.
7. Add gateway and credentials (if required).
8. Click on "Apply".

Important Points:

- Available in Power BI Pro and Premium.
- Maximum 8 refreshes per day (Pro).
- Maximum 48 refreshes per day (Premium).
- Requires a Gateway for on-premises data sources.
- Refresh will fail if server is offline.
- You can view refresh history in dataset settings.
- Use incremental refresh for large datasets.

71. Dataflow

A Dataflow is a cloud-based ETL (Extract, Transform, Load) feature in Power BI Service. It allows you to ingest, transform, and load data into Power BI workspaces so multiple reports and models can reuse the same cleaned data source without repeating transformations.

72. What is Data Modeling in Power BI?

Data modeling is the process of organizing and connecting data from different tables using relationships. It helps in creating a proper structure for data analysis. A good data model ensures accurate results, faster report performance, and simplifies complex calculations. In Power BI, we use data modeling to define relationships and set the correct model structure (like Star Schema).

Key Points:

- Data model is created in the Model view.
- Connects different tables using keys.
- Helps in accurate and faster data analysis.
- Uses fact tables and dimension tables.
- Proper data modeling improves performance.

Benefits of Data Modeling:

- Combines data from multiple sources.
- Reduces data redundancy.
- Improves query performance.
- Makes DAX calculations easier.
- Provides a clear and logical structure.
- Helps in better data analysis and reporting.

73. Star Schema vs Snowflake Schema

Star Schema	Snowflake Schema
• Consists of one central Fact Table and multiple Dimension Tables. • Dimension tables are directly connected to Fact table. • Data is not normalized (denormalized). • Better performance in Power BI. • Simple and preferred structure for reporting.	• Dimension tables are normalized into multiple related tables. • Dimension tables are split into sub-dimensions. • More complex structure. • May have slightly lower performance. • Used when data is very large and highly normalized. • Not commonly used in simple Power BI reporting.

74. Components of a Data Model

Component	Description	Example	Type
Fact Table	Contains numeric data (measures)	Sales, Orders, Revenue	Fact
Dimension Table	Contains description data (attributes)	Product, Customer, Date	Dimension
Relationship	Connects Fact and Dimension tables	ProductID, CustomerID	One-to-Many (1:*)
Key	Unique identifier for each record	ProductID, DateID	Primary Key
Cardinality	Defines relationship type	One-to-Many, Many-to-Many	Cardinality
Cross Filter Direction	Defines filter flow between tables	Single (default), Both	Filter Direction

75. Example: Star Schema in Power BI

A central **Fact Table (Sales)** surrounded by three **Dimension Tables**:

- **Customer Dimension Table:** CustomerID (PK), CustomerName, City, Country → *(1:*)* → **Sales Fact Table:** SalesID, CustomerID (FK), ProductID (FK), DateID (FK), Quantity, Amount
- **Product Dimension Table:** ProductID (PK), ProductName, Category, Price → *(1:*)* → **Sales Fact Table**
- **Date Dimension Table:** DateID (PK), Date, Month, Quarter, Year → *(1:*)* → **Sales Fact Table**

> **Key Points in Star Schema:** Fact table is in the center; Dimension tables are around the fact table; Relationships are 1-to-many (1:*); Dimension tables filter the fact table; Use clear and meaningful table names.

76. Best Practices for Data Modeling in Power BI

Do's:

- Use a star schema structure.
- Keep table names and columns clean and unique.
- Use proper data types.
- Hide unused columns from report view.

Don'ts:

- Avoid many-to-many relationships (unless required).
- Avoid complex snowflake schemas for simple reports.
- Avoid bi-directional filtering (use only when needed).
- Do not create unnecessary relationships.

77. Relationships in Power BI

Relationships in Power BI are used to connect tables based on common columns. They help in combining data from multiple tables for accurate calculation and analysis.

Key Points:

- Relationships are created in Model view.
- Tables are connected using common columns (keys).
- Helps in data analysis and reporting.
- Supports automatic filter propagation.
- Can be one-to-one, one-to-many or many-to-many.

Types of Relationships:

- One-to-One (1:1)
- One-to-Many (1:*)
- Many-to-Many (*:*)
- Cross Filter Direction (Single or Both)
- Active and Inactive Relationships
- Cardinality and its importance

78. Types of Relationships

Type	Description	Example
One-to-One (1:1)	Each record in one table matches one record in another table.	Employee and Employee Details
One-to-Many (1:*)	One record in the first table can be related to multiple records in the second table.	Customer → Orders
Many-to-Many (*:*)	Multiple records in one table can be related to multiple records in another table.	Students ↔ Courses

79. Cross Filter Direction

Single Direction (Default)	Both Direction
• Filter flows from one table to another (1 → *). • Usually from dimension to fact table. • Better performance and recommended. • Most common in star schema. *Filter Direction: Customer (Dim) → Sales (Fact)*	• Filter flows in both directions (* ↔ *). • Useful in many-to-many relationships or complex calculations. • May impact performance. • Use only when required. *Filter Direction: Customer (Dim) ↔ Sales (Fact)*

80. Example: Relationships in a Sales Data Model

A classic analytical relational schema where Customer, Product, and Date dimension tables filter a central Sales Fact table via One-to-Many (1:*) single-direction relationships:

- `Customer[CustomerID] (1) → (*) Sales[CustomerID]`
- `Product[ProductID] (1) → (*) Sales[ProductID]`
- `Date[DateID] (1) → (*) Sales[DateID]`

81. Best Practices for Creating Relationships

Do's:

- Use clear and meaningful column names.
- Ensure data in the key column is clean and unique.
- Use single direction filter where possible.
- Use star schema for better performance.
- Keep dimension tables unique (no duplicate keys).

Don'ts:

- Do not create many-to-many relationships unless required.
- Do not use both-direction filter without a valid reason.
- Do not create active relationships between duplicate values.
- Do not create unnecessary relationships.

82. What is Power Query?

Power Query is a data preparation and data transformation tool in Power BI used to connect, clean, transform and prepare data from various sources before loading it into the data model. It provides a visual interface to perform complex data transformations easily and automatically generates M code behind each applied step.

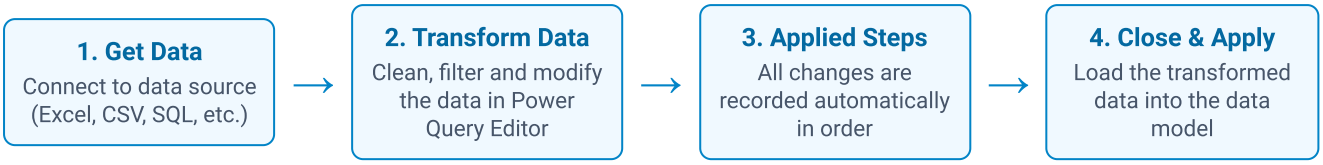
Key Points:

- Used for data cleaning and transformation.
- Helps in cleaning and shaping raw data.
- Automates repeated ETL processes.
- Easy to use graphical interface.
- Keeps a record of all applied steps.
- Generates M code automatically.

Common Data Sources:

- Excel files
- CSV / Text files
- SQL Server / Databases
- Web pages / APIs
- Cloud sources (Azure, MySQL, etc.)
- Folders (Multiple files)

83. Steps to Use Power Query in Power BI



84. Common Transformations in Power Query

Transformation	Description	Use Case	Example
Remove Rows	Delete unwanted rows	Remove blank or error rows	Remove null values
Choose Columns	Select necessary columns	Keep only required data	Remove extra columns
Change Data Type	Convert column data type	Set text, number, date, etc.	Text to Date
Rename Columns	Change column names	Make names meaningful	`Cust_ID` to `CustomerID`
Filter Rows	Keep specific data	Filter by condition	Show sales > 2000
Sort Data	Arrange data in order	Sort ascending/descending	Sort by Date
Split Column	Split a column into multiple	Separate combined data	FullName to First/Last
Replace Values	Replace specific values	Handle missing or wrong data	Replace "NA" with 0

85. Merge Queries vs. Append Queries

Merge Queries (Join Tables)	Append Queries (Combine Rows)
• Combines columns from two or more tables. • Similar to SQL JOIN. • Based on a common column in different tables but related. • Types: Left Join, Right Join, Inner Join, Full Outer Join. • *Customer Table + Sales Table → Merged Table with joined columns*	• Combines rows from two or more tables. • Tables must have the same columns. • Similar to SQL UNION ALL. • Adds rows from one table to the bottom of another table. • Commonly used to combine data from multiple months or multiple files. • *Jan Sales + Feb Sales → Appended Unified Table*

86. Pivot and Unpivot Columns

Pivot Column	Unpivot Column
• Converts unique values from a column into multiple columns. • Useful when data is in a long format.	• Converts multiple columns into rows. • Useful when data is in a wide format. • Commonly used to transform multiple month columns into normalized rows.

"Learn Today, Build Better Reports Tomorrow."

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